$$\iint_R f(x,y)\,dA = \iint_E F(u,v)\left|\frac{\partial(x,y)}{\partial(u,v)}\right| du\,dv$$

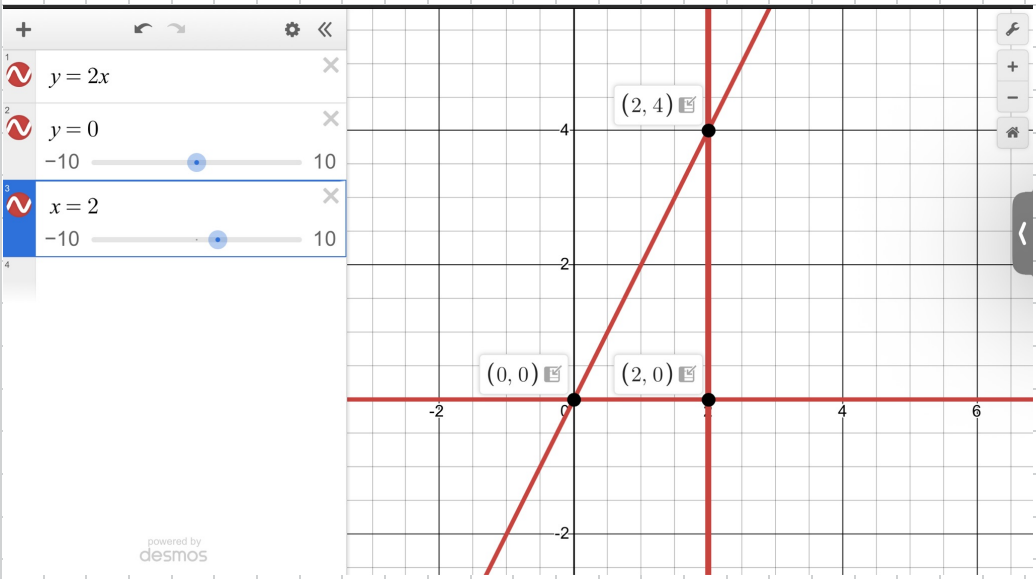
Se recorre R y E en sentido antihorario ↺

Ejemplo: Usar el cambio de variable indicado para expresar la integral como una integral doble sobre una región S en el plano uv.

(No evaluar la integral)

$$\iint_R (y-x)\,dx\,dy \;;$$

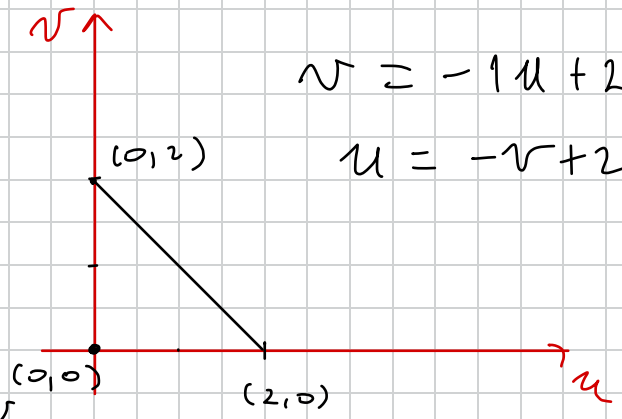
donde R es la región acotada por $y=2x$, $y=0$, $x=2$; $x=u+v$; $y=2v$.



$$R = \{(x,y) \in \mathbb{R}^2 \mid (0,0) \longrightarrow (2,0) \longrightarrow (2,4)\}$$

Forman un triángulo con los vértices.

x	y	u	v
0	0	0	0
2	0	2	0
2	4	0	2



$v = -1u + 2$

$u = -v + 2$

$x = u + v \quad ; \quad y = 2v.$

$x = u + \frac{y}{2} \quad ; \quad v = \frac{y}{2} \quad ; \quad u = x - \frac{y}{2}$

$$J = \frac{\partial(x,y)}{\partial(u,v)} = \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} = (1)(2) - (1)(0)$$

$$\iint_R (y - x)\,dx\,dy = \int_0^2 \int_0^{-v+2} (v - u)(2)\,du\,dv$$

$$= 2\int_0^2 \left(vu - \frac{u^2}{2} \right)\Big|_0^{-v+2} dv$$

$$= 2\int_0^2 \left(-v^2 + 2v - \frac{(-v+2)^2}{2} \right) dv$$

$t = -v + 2$

$dt = -dv$

$$= 2\left(-\frac{v^3}{3} + v^2 + \frac{(-v+2)^3}{6} \right)\Big|_0^2$$

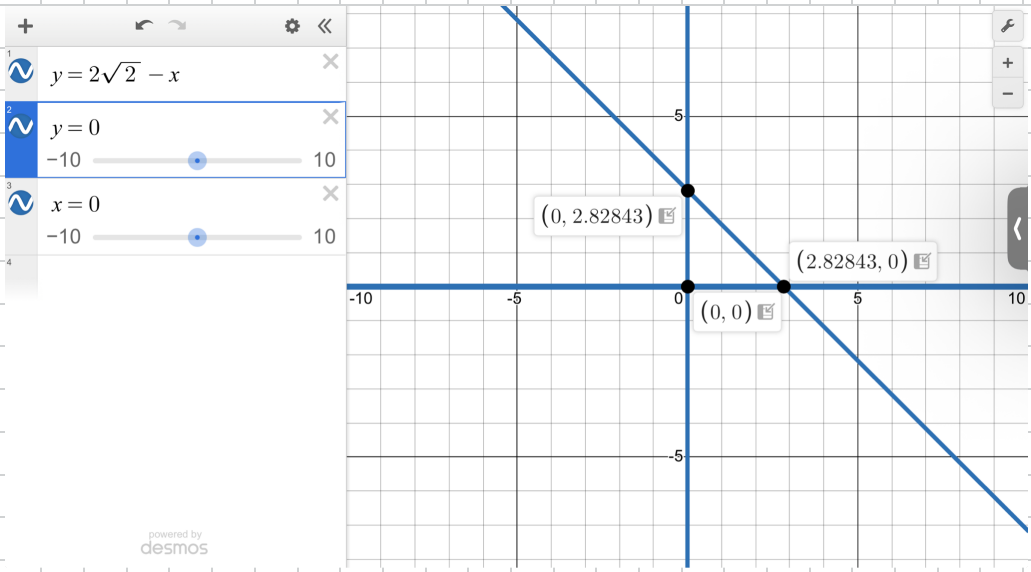
$$= 2\left(-\frac{8}{3} + 4 - \frac{8}{6} \right) = 2\left(-\frac{24}{6} + 4 \right) = 0$$

Ejemplo: Usar el cambio de variable indicado para expresar la integral como una integral doble sobre una región S en el plano uv.

(No evaluar la integral)

$$\iint_R xy\,dx\,dy \quad ;$$

donde R es la región acotada por $y = 2\sqrt{2} - x$, $x = 0$, $y = 0$; $x = u^2 - v^2$; $y = 2uv$.



$$J = \frac{\partial(x,y)}{\partial(u,v)} = \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{bmatrix} = \begin{bmatrix} 2u & -2v \\ 2v & 2u \end{bmatrix} = 4u^2 + 4v^2$$

$x = u^2 - v^2$; $y = 2uv$.

$u^2 + v^2 = \sqrt{x^2+y^2}$

$u^2 - v^2 = x$

$2u^2 = \sqrt{x^2+y^2} + x$

$u = \sqrt{\frac{\sqrt{x^2+y^2} + x}{2}}$

$2v^2 = \sqrt{x^2+y^2} - x$

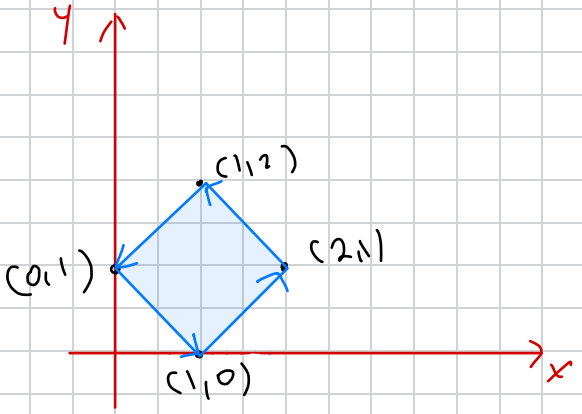
$v = \sqrt{\frac{\sqrt{x^2+y^2} - x}{2}}$

$$\iint_S (u^2 - v^2)\, 2uv\, (4u^2 + 4v^2)\, du\, dv$$

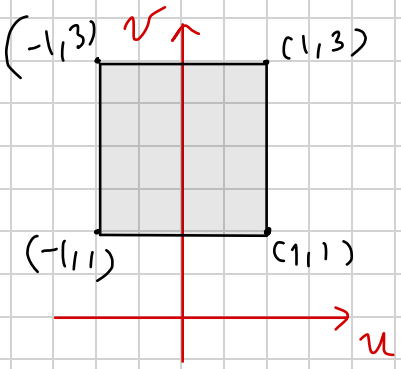
Ejemplo: Evaluar la integral haciendo el cambio de variable indicado:

$$\iint_R (x-y)^2 \cos^2(x+y)\, dx\, dy \;;$$

R es la región acotada por los vértices $(0,1)$, $(1,2)$, $(2,1)$, $(1,0)$. $u = x - y$; $v = x + y$



x	y	u	v
1	0	1	1
2	1	1	3
1	2	−1	3
0	1	−1	1



$$\int_1^3 \int_{-1}^{1} u^2 \cos^2 v\, du\, dv$$

$$\cos^2(v) = \frac{1}{2}(1 + \cos(2v))$$

$$= \frac{1}{2}\left(v + \frac{\operatorname{sen}(2v)}{2}\right)$$

$$\int_1^3 \frac{u^3}{3}\Big|_{-1}^{1} (\cos^2(v)\, dv = \frac{2}{3}\int_1^3 \frac{1}{2}(1 + \cos(2v))\, dv$$

$$= \frac{2}{3}\cdot\frac{1}{2}\left(v + \frac{\operatorname{sen}(2v)}{2}\right)\Big|_1^3 = \frac{1}{3}\left(2 + \operatorname{sen}(6) - \operatorname{sen}(2)\right)$$